



Investigating Climate Health: New Research on Air Pollution and Extreme Heat

Abstract

In 2026, researchers at Harvard T.H. Chan School of Public Health launched three new projects funded by Wellcome, a London-based health research foundation. One studies how fossil fuel air pollution drives disease, another measures the health benefits of fighting climate change in India, and a third tracks how extreme heat affects women who work in India's informal economy. Together they show how climate health research moves from the lab to communities to policy.

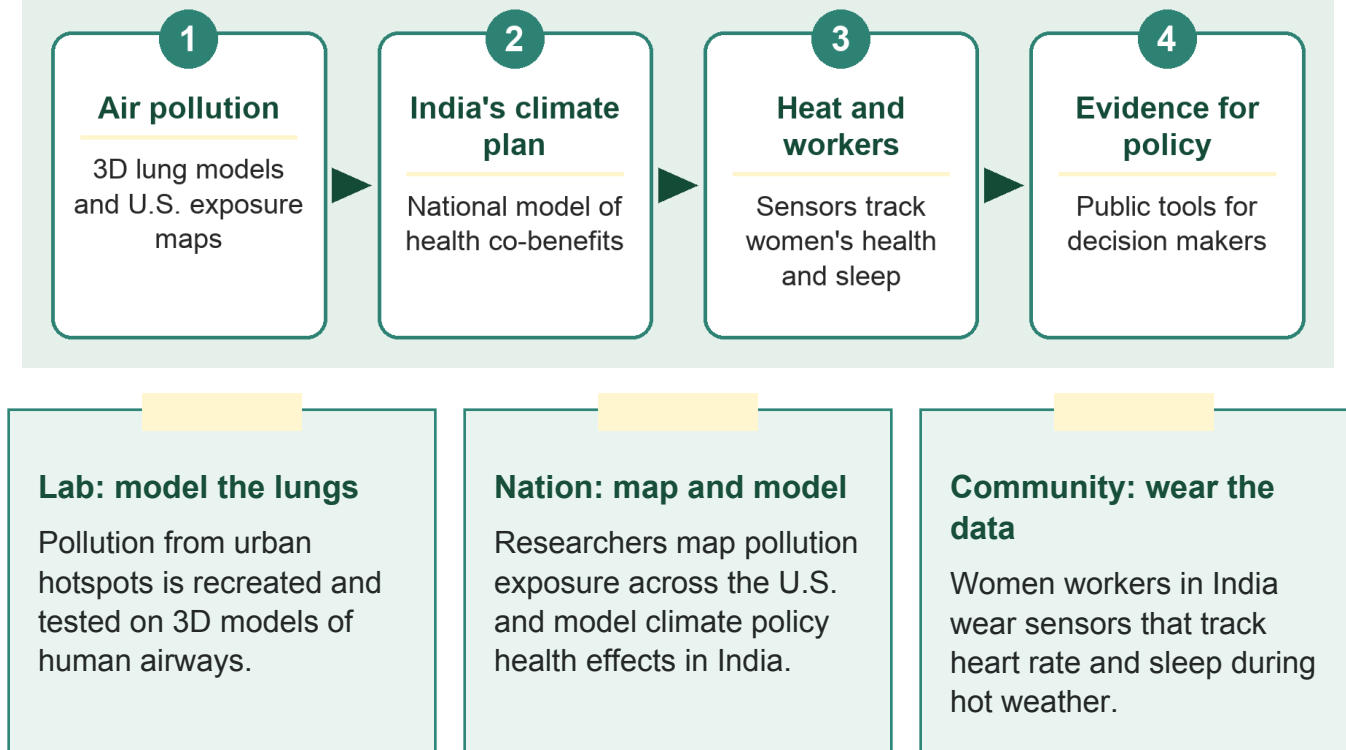
Introduction

Burning fossil fuels warms the planet and pollutes the air, and both harm human health. But which pollutants cause which diseases, how much do climate solutions improve health, and how does heat affect the people who have the least protection? These are the questions behind three new research projects supported by multi-million-dollar grants from Wellcome. The projects are based in the United States and India, the world's third-largest carbon dioxide emitter. Each one uses different methods, from 3D models of human airways to wearable sensors, but they share a goal: producing evidence that decision makers can use to protect health in a warming world.

How Do Scientists Study This?

The three projects use very different tools. In the air pollution project, led by Mary Berlik Rice with a \$4 million award, researchers collect source-specific fine particle pollution (PM2.5) from urban hotspots, recreate the polluted air in the lab, and expose 3D models that behave like real lung airways. They also map how much pollution people breathe across the U.S. and link exposure levels to disease. In India, Gaurab Basu's team (\$2.5 million) will build the country's first national climate health model with a government think tank. In the Community HATS study (\$3 million), women workers wear devices tracking heart rate and sleep for a year while researchers measure heat and humidity in their homes and workplaces.

Three projects, three scales



What Did They Discover?

These projects are just beginning, but they are designed to produce practical tools. The air pollution team will create a public interactive tool to help U.S. policymakers judge the effectiveness of climate mitigation measures. The India climate model will forecast how climate policies affect health, energy security, and economic growth through 2070, supporting India's goal of net-zero emissions by 2070; the team will also study electricity-reduction programs in Chennai and Delhi and clean cooking in coal mining communities in Jharkhand. Community HATS, co-designed by the women workers themselves, is poised to become the largest ever long-term study of worker heat stress, collecting about 9.5 million observations from more than 800 women.

The Wellcome projects

\$9.5M

in total Wellcome grants: \$4M, \$2.5M, and \$3M

800+

women workers in the Community HATS study

9.5M

distinct observations expected from Community HATS

Unlike earlier heat studies on athletes or soldiers in labs, Community HATS measures heat in the real lives of low-income workers, who help collect and share the data.

Takeaways

From cells to policy

Research spans lab models of airways to national tools that guide climate policy.

Climate action is health action

Cutting emissions can bring health and economic co-benefits, as India's model will show.

Measure real lives

Wearable sensors reveal how heat affects the health of workers with the least protection.

Key Terms

PM2.5

Fine particulate matter less than 2.5 micrometers wide. It can travel deep into the lungs and bloodstream.

Climate mitigation

Actions that reduce greenhouse gas emissions to slow climate change, such as using cleaner energy.

Net-zero

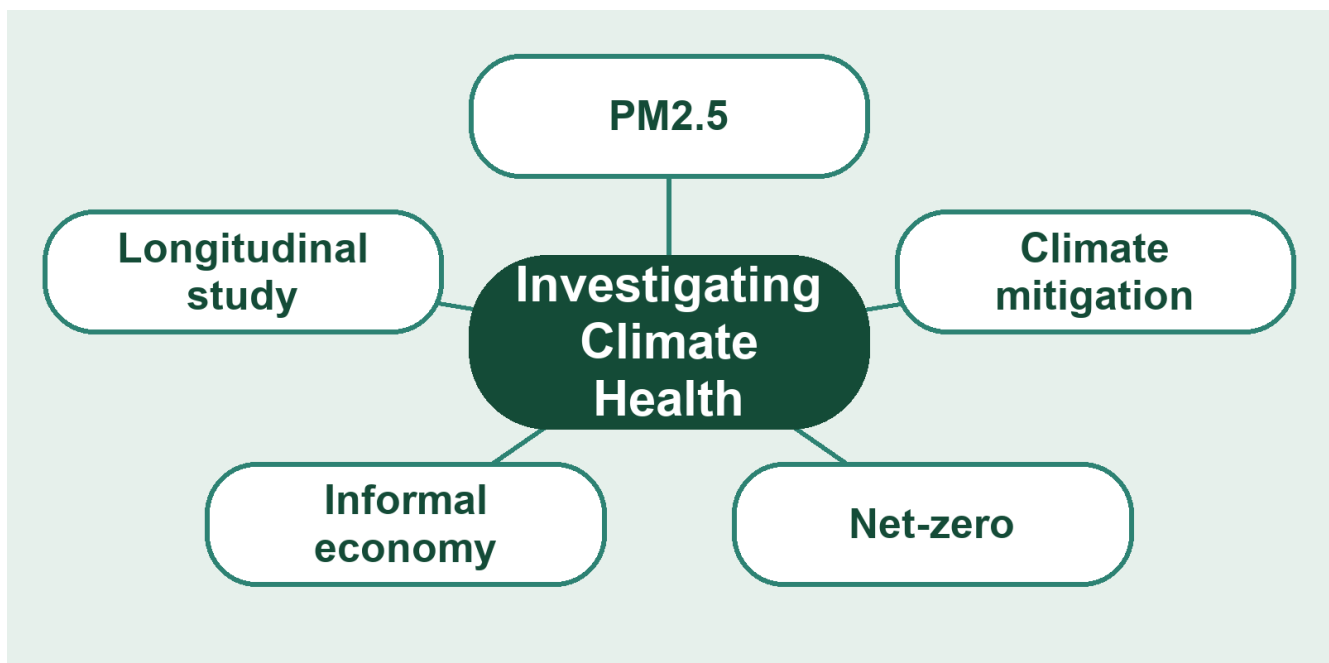
Balancing the greenhouse gases a country emits with the amount it removes from the atmosphere.

Informal economy

Work that is not officially regulated or protected, such as street vending, home-based crafts, and small farming.

Longitudinal study

Research that follows the same people over a long period to see how things change.



Activity

Three Scales of Research

Students compare the three Wellcome-funded projects and design their own study at one scale.

Materials: Comparison chart: lab, nation, community; Project summary cards; Markers

1. Fill in the chart for each project: the question, the method, the data collected, and the intended users.
2. Discuss what each scale of research can reveal that the others cannot.
3. Choose one scale and design a small study about heat or air quality in your own community.
4. Explain who would use your results and how the study could protect health.
5. Reflect: why does it matter that Community HATS participants co-designed the study?

References

Brownstein, M. (2026). Researchers launch new climate health projects with Wellcome support. Harvard T.H. Chan School of Public Health.

<https://hsph.harvard.edu/news/researchers-launch-new-climate-health-projects-with-wellcome-support/>